TESSELLATION

In its most basic form, tessellation is a method of breaking down polygons into finer pieces. For example, if you take a square and cut it across its diagonal, you've "tessellated" this square into two triangles. By itself, tessellation does little to improve realism. For example, in a game, it doesn't really matter if a square is rendered as two triangles or two thousand triangles—tessellation only improves realism if the new triangles are put to use in depicting new information, which is done by displacement mapping. Displacement mapping stores the height of each vertice and thus we get a 3D layer. Tessellation has been around for some time but for proper realism there had to be a lot of vertices to be present. Earlier, GPUs in general lacked computational power which is now easily available. With DirectX 11, Tessellation became more mainstream and optimised. NVIDIA GPUs from the GeForce 400 series have proven to show upto 100% enhanced output in comparison to it's competitors. With the newer generation of GPUs coming up, gamers are in for a great treat.

3D VISION 2

Three years since their 3D Vision technology premiered NVIDIA recently launched the next iteration, 3D Vision 2. An upgraded battery(approximately 60 hours), compatibility with previous generation technology, larger glasses(approximately 20% larger) and better integration are amongst the highlights of the new release. Bigger glasses mean a bigger frame which in turn translates to

lesser unwanted light creeping in from the periphery. The glasses have a range of up to 15 feet. Also, the glasses come with a USB 2.0 micro port. A few monitor manufacturers have even integrated the IR transmitters into a few of their models. The glasses remain to be shutter based which have an inherent flaw of blocking too much light and thus the images appear way too dark. 3D Vision too comes with a new technology called 3D LightBoost which tries to mitigate the issue of dark playback. However, for proper implementation of the technology requires modification of the monitor's hardware. With over 600 titles utilising this technology and many more adopting it, an endless stream of 3D entertainment awaits.

ANISOTROPIC FILTERING

Anisotropy is the quality of possessing dissimilar coordinate values in a space, which applies to any texture not displayed as absolutely perpendicular to the camera. Anisotropic filtering improves the clarity and crispness of textured objects in games.

To simultaneously preserve performance and image quality, mipmaps are used; mipmaps are duplicates of a master texture that have been pre-rendered at lower resolutions which a graphics engine can call when a corresponding surface is a specific distance from the camera.

With proper filtering, the use of multiple mipmap levels in a scene can have no discernable impact on its appearance while greatly optimizing performance.

Bilinear filtering serves as the default, being the simplest and computationally cheapest form of texture filtering available due to its simple approach:to calculate a texel's final color, four texel samples are taken from the mipmap defined by the graphics engine at the approximate point where the target texel exists on-screen, which will appear as a combined result of those samples' color data.

Anisotropic filtering can be controlled through the NVIDIA Control Panel within the 3D Settings section, however for the best performance and compatibility NVIDIA recommends that users set this to be controlled by the application.